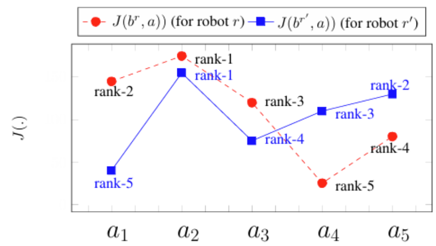
Endsem, CSE667 (DMMRS) Full marks: 40 25.04.2026

Your answers should be i) precise, ii) to-the-point, and iii) with legible handwriting.

1. Find the MSNE of the game in Figure 1(a). You should adapt an efficient approach. What is the *support* of the mixed strategy? 5+2

		P-2		
		X	Y	Z
P-1	A	5, 1	6, 2	1, 3
	B	2, 3	5, 4	6, 0
	C	3, 2	3, 0	0, 4

(a)



(b)

Figure 1: (a) For Question 1 , and (b) for Question 2

Solution:

1/

		P2		
		X	Y	Z
	A	5, 1	6, 2	1, 3
P1	B	2, 3	5, 4	6, 0
	C	3, 2	3, 0	0, 4

Strategy C of P1 can be eliminated as strategy A strongly dominates C.

	X	Y	Z
A	5, 1	6, 2	1, 3
B	2, 3	5, 4	6, 0

Strategy X of P2 can be eliminated as strategy Y strongly dominates X.

	Y (y)	Z $(1-y)$
A (x)	6, 2	1, 3
B $(1-x)$	5, 4	6, 0

As per the necessary and sufficient condition for MSNE,

$$u_1(A) = u_1(B) \quad \text{—— } (1)$$

and

$$u_2(Y) = u_2(Z) \quad \text{—— } (2)$$

From (1), $6y + 1\cdot(1-y) = 5y + 6(1-y)$

$$\Rightarrow y = \frac{5}{6} \quad ; \quad 1-y = \frac{1}{6}$$

From (2), $2x + 4\cdot(1-x) = 3\cdot x + 0\cdot(1-x)$

$$\Rightarrow x = \frac{4}{5} \quad ; \quad 1-x = \frac{1}{5}$$

MSNE: $\left[\left(\frac{4}{5}, \frac{1}{5}, 0\right) ; \left(0, \frac{5}{6}, \frac{1}{6}\right)\right]$

Support of the mixed strategy is

$$\{A, B\} \times \{Y, Z\}$$

Evaluation:

- Elimination of two strategies: $1 \times 2 = 2$
- Equating the payoffs (eq.1 and eq2 above): 1
- Solving for the mixed strategies: $1 \times 2 = 2$
- Finding support of the mixed strategy: 2

2. We studied multi-robot action consistency (MR-AC).

   i) For an arbitrary robot r, write the mathematical expression for objective function for a look-ahead of 1. 1.5

   ii) Refer Figure 1(b). Consider robots r and r'. If $bel_k^r \neq bel_k^{r'}$ then, in general, what effect does it have on action selection for a given sequence of actions ? 1.5

   iii) Refer Figure 1(b). If each possible observation of the robots has a likelihood of occurrence then suggest an approach to solve MR-AC with a partial set of observations (instead of a full set). 3

3. A robot uses a range sensor that can measure ranges from $0m$ and $3m$. Assume that actual ranges are distributed *uniformly* in this interval. Unfortunately, the sensor can be faulty. When the sensor is faulty, it constantly outputs a range $< 1m$, regardless of the actual range. The prior probability for a sensor to be faulty is p .

   Consider that sensor state is binary: faulty (F) and non-faulty ($\neg F$); and possible observations are $z < 1$ and $z \not< 1$. Answer the following:

   i) Find the measurement model. 2

   ii) Suppose the robot queried its sensor N times independently, and every single time the measurement value is $< 1m$. Find $\overline{bel}_N(F)$ and $bel_N(F)$. 4

Solution:

3/ Sensor state:

Faulty (F) , Not faulty ($\neg F$)

Possible observations (relevant to the question):

$z < 1m$, $z \geqslant 1m$

(i) Measurement model:

$$p(z < 1m \mid F) = 1$$
$$p(z \geqslant 1m \mid F) = 0$$
$$p(z < 1m \mid \neg F) = \frac{1}{3}$$
$$p(z \geqslant 1m \mid \neg F) = \frac{2}{3}$$

(ii) $\overline{bel}_N(F)$ and $bel_N(F)$:

Robot queried its sensor N times independently with $z < 1m$ each.

S = sequence of N observations with $z < 1m$ each.

$$p(S \mid F) = 1^N = 1$$
$$p(S \mid \neg F) = \left(\frac{1}{3}\right)^N$$
$$p(F) = p \quad , \quad p(\neg F) = 1 - p \quad [\text{Given}]$$

$$\mathrm{bel}_N(F) = p(F \mid S)$$

$$= \frac{p(S \mid F) \cdot p(F)}{p(S \mid F) \cdot p(F) + p(S \mid \neg F) \cdot p(\neg F)}$$

$$= \frac{1 \cdot p}{1 \cdot p + \left(\frac{1}{3}\right)^N \cdot (1-p)}$$

$$= \frac{p}{p + (1-p) \cdot \frac{1}{3^N}}$$

$$\overline{\mathrm{bel}}_N(F) = \mathrm{bel}_{N-1}(F)$$

$$= \frac{p}{p + (1-p) \cdot \frac{1}{3^{N-1}}}$$

Evaluation:

- Q3(i): $0.5 \times 4 = 2$
- Q3(ii) – find $P(S \mid F)$ and $P(S \mid \neg F)$: $0.5 \times 2 = 1$
- Q3(ii) – find $bel_N(F)$: 2
- Q3(ii) – find $\overline{bel}_N(F)$: 1

4. Write down the Monte Carlo Localization algorithm which applies the concept of particle filters. 5

Solution:

Algorithm MCL($\mathcal{X}_{t-1}, u_t, z_t, m$):
  $\bar{\mathcal{X}}_t = \mathcal{X}_t = \emptyset$
  for $m = 1$ *to* M *do*
    $x_t^{[m]} =$ **sample_motion_model**$(u_t, x_{t-1}^{[m]})$
    $w_t^{[m]} =$ **measurement_model**$(z_t, x_t^{[m]}, m)$
    $\bar{\mathcal{X}}_t = \bar{\mathcal{X}}_t + \langle x_t^{[m]}, w_t^{[m]} \rangle$
  endfor
  for $m = 1$ *to* M *do*
    draw i *with probability* $\propto w_t^{[i]}$
    add $x_t^{[i]}$ *to* $\mathcal{X}_t$
  endfor
  return $\mathcal{X}_t$

Table 8.2 MCL, or Monte Carlo Localization, a localization algorithm based on particle filters.

Evaluation:

- Input parameters: 1
- Output: 1
- Motion model: 1
- Measurement model: 1
- Resampling: 1

5. (a) Refer Figure 2(a). Consider a simplified global localization in the following grid-style environment. A robot will be placed at the random position facing *North*. A open-loop localization strategy contains a sequence of following commands:

- Action L: Turn left 90 degrees
- Action R: Turn right 90 degrees
- Action M: Move forward until you hit an obstacle

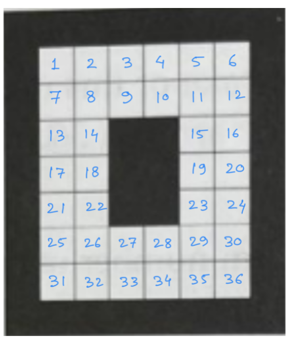
The robot moves in a way that minimizes location uncertainty. Answer the following:

i) Calculate the initial belief bel_0 of the robot. 1

ii) Calculate bel_1 of the robot. 1.5

iii) Calculate bel_2 of the robot. 1.5

(b) In *any* open-loop localization problem where the robot wants to reduce localization uncertainty, when should the robot stop moving ? 3



(a)

cell no.	type (actual)	measurement sequence						
cell-1	occupied	1	1	0	1	0	1	1
cell-2	occupied	0	1	1	1	0	0	1
cell-3	free	0	0	0	0	0	0	0
cell-4	free	1	0	0	1	0	0	0

(b)

Figure 2: (a) For Question 5 , and (b) for Question 6

Solution:

5. (a)

(i) Initially, all the obstacle-free locations are equally likely. Therefore,

$$bel_0(x=i) = \frac{1}{36} \quad \text{for } 1 \leq i \leq 36$$

For minimizing location uncertainty, the robot takes action sequence: M, RM, RM

(ii) So, at time step 1, belief is

$$bel_1(x=i) = \frac{1}{8} \quad \text{for } i = 1 \text{ to } 6 \text{ and } 27, 28$$

$$= 0 \quad \text{elsewhere.}$$

(iii) At time step 2, belief is

$$bel_2(x=i) = 1 \quad \text{for } i = 6 \text{ or } 30$$

$$= 0 \quad \text{elsewhere.}$$

5. (b) For reducing localization uncertainty, the robot navigates through the workspace and increases likelihood of some locations while reduces likelihood to zero of some other locations.

The robot should stop moving when in the reduced state space $X' \subseteq X$ the following occurs:

(i) any pair (x_i, x_j), where $x_i, x_j \in X'$, are indistinguishable,

OR

(ii) any pair (x_i, x_j), where $x_i, x_j \in X'$, are distinguishable.

OR

(iii) X' contains a single element (singleton)

Evaluation:

- Q5(a) Optimal action sequence: 1
- Q5(a)(i): 1
- Q5(a)(ii): $0.5 + 0.5$
- Q5(a)(iii): $0.5 + 0.5$
- Q5(b): $1 \times 3 = 3$

6. In Figure 2(b), binary occupancy measurements for the following four cells are given. Consider that m_i is a binary occupancy variable, and z is a binary measurement variable. Design a measurement model $\{p(z \mid m_i)\}$. 5

Solution:

6/ Occupied cells (cell 1 and 2)

$z=1$: $5+4=9$

$z=0$: $2+3=5$

Free cells (cell 3 and 4)

$z=1$: 2

$z=0$: 12

$P(z=1 \mid m_i=1) = \frac{9}{14}$ ✓

$P(z=0 \mid m_i=1) = \frac{5}{14}$ ✓

$P(z=1 \mid m_i=0) = \frac{2}{14} = \frac{1}{7}$ ✓

$P(z=0 \mid m_i=0) = \frac{12}{14} = \frac{6}{7}$ ✓

Evaluation:

- $z = 0$ or 1 is defined: 1
- $m_i = 0$ or 1 is defined: 1
- Correct answer: 1.5 + 1.5

7. Consider a world with four possible robot locations: $X = \{x_1, x_2, x_3, x_4\}$. Initially, we draw N samples uniformly from among those locations. Let Z be a Boolean sensor

variable with the following conditional probabilities:

$$p(z \mid x_1) = 0.8,\ p(z \mid x_2) = 0.4,\ p(z \mid x_3) = 0.1 \text{ and } p(z \mid x_4) = 0.1.$$

Assume that we only generate one new sample in the resampling process, regardless of N. This sample might correspond to any of the four locations in X. Thus, the sampling process defines a probability distribution over X.

What is the resulting probability distribution over X for this new sample for i) $N = 10$ and ii) $N = \infty$? 2+2

Solution:

7/ Resampling is done based on the prior distribution and the observation [as per observation model]

Initially, $p(x_1) = p(x_2) = p(x_3) = p(x_4) = \frac{1}{4}$

[given in the question]

$p(z|x_1) = 0.8$, $p(z|x_2) = 0.4$, $p(z|x_3) = 0.1$
and $p(z|x_4) = 0.1$ [observation model].

Each particle at location x_i gets $w_i \propto p(z|x_i)$

Normalized posterior weight $w_i = \eta \, p(z|x_i) \cdot p(x_i)$

Thus, the normalized distribution is:

$$p(x_i|z) = \eta \, p(z|x_i) \cdot p(x_i)$$

So, $p(x_1) = \frac{0.8}{1.4} = 0.57$ ✓

$p(x_2) = \frac{0.4}{1.4} = 0.285$ ✓

$p(x_3) = \frac{0.1}{1.4} = 0.07$ ✓

$p(x_4) = \frac{0.1}{1.4} = 0.07$ ✓

Evaluation:

- Q7(i) Particle weight w_i: 0.5
- Q7(i) Normalization: 0.5
- Q7(i) Correct probability distribution: 1
- Q7(ii) *No change for* $N = \infty$ *compared to* $N = 5$: 2